

NIKIL R

AI Engineer | LLM Systems | Backend Developer

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PROFESSIONAL SUMMARY

Final-year Information Science undergraduate targeting AI Engineer roles. Built 3 production-grade AI systems — a hybrid RAG pipeline, a distributed 6-agent content orchestration system, and an autonomous LangGraph recruitment swarm with semantic memory and self-healing validation. Strong in agentic system design, async backend engineering, and end-to-end LLM integration.

EDUCATION

Bachelor of Engineering (B.E.) in Information Science

Sambhram Institute of Technology2022 – 2026

CGPA: 8.08 / 10

Pre-University (12th Grade)

ASC PU College2022

Percentage: 67%

SSLC (10th Grade)

New Oxford High School2020

Percentage: 74%

TECHNICAL SKILLS

Languages

Python

SQL

JavaScript

AI Engineering

LangChain

LangGraph

RAG

Agentic AI

Multi-Agent Systems

Semantic Search

ChromaDB

Embeddings

Vector Search

Knowledge Retrieval

Backend & APIs

FastAPI

REST APIs

Redis

WebSockets

Celery

Databases

PostgreSQL

MySQL

Cloud & Deployment

AWS

Docker

Nginx

Tools

Git

GitHub

Postman

LangSmith

EXPERIENCE

AI & Full Stack Developer Intern

Nano Stream TechnologiesJan 2026 – May 2026

- Built frontend modules for a **production-grade restaurant POS system** using React and TypeScript, covering billing, analytics, and order management.
- Deployed and maintained the application on **AWS EC2**, ensuring production accessibility and reliable real-time workflow handling.
- Resolved frontend integration, state management, and deployment issues, improving overall application stability and performance.

PROJECTS

LegalFinance RAG System

Python, FastAPI, ChromaDB, Semantic Search, RAG

- Built a production RAG pipeline processing **300+ legal and financial documents** with LLM-based query answering.
- Implemented hybrid retrieval using embeddings and vector search, improving response relevance by **~30%**.
- Deployed FastAPI endpoints enabling real-time document querying with **sub-second response times (400–800ms)**.

Content Agent System — Multi-Agent Orchestration

Python, FastAPI, LangChain, Celery, PostgreSQL, React

- Architected a distributed swarm of **6 specialized agents** for autonomous content planning, generation, and evaluation.
- Engineered async orchestration using **FastAPI, Celery, and Redis** for scalable, non-blocking multi-step LLM pipelines.
- Reduced hallucination rate by **85.6%** via shared-state coordination, critic-optimizer loops, and real-time WebSocket monitoring.

HireIQ — Autonomous Multi-Agent Research Swarm

Python, LangGraph, FastAPI, ChromaDB, Tavily API, React

- Engineered a **6-agent LangGraph swarm** with recursive backtracking and a self-healing Judge node for autonomous validation.
- Implemented semantic memory via **ChromaDB** embeddings, reducing repeated query latency from **42.5s to 100ms**.
- Achieved full observability via **LangSmith** tracing and validated agent workflows with a **Playwright + Pytest** test suite.